



The Bankability-by-Design Framework

Making climate adaptation investable, from idea to investment case



Co-funded by
the European Union

Bankability-by-Design

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Why adaptation projects **don't** get financed

The capital is there; the investable projects are not. Barriers identified in CLIMINVEST's analysis (D2.1, D2.2):



Value is avoided loss

Benefits appear as damage that does not happen, not as revenue.



No bankable payer

Those who benefit are rarely a counterparty a lender can underwrite.



Co-benefits are invisible

Social, ecological and resilience value goes unmeasured, so it is priced at zero.



Translation burden

Engineers, authorities and investors assess the same project in different units.



Institutional readiness

Smaller public bodies lack structuring, reporting and O&M capacity.



High transaction cost

Every project is assessed from scratch; due diligence outweighs small tickets.

The gap is not capital. It is investable projects.

Where the value sits, and where the cash doesn't

Pilot 1, SOLVATE (PV over an irrigation canal, Thessaly): who captures each value stream?

VALUE STREAM	TYPE · BENEFICIARY	CASH?
Avoided evaporative water loss	Public good · irrigation users, river basin	No
Irrigation continuity under drought	Avoided loss · farmers, regional economy	No
Local O&M employment	Social · local labour market	No
CO ₂ avoided	Environmental	Depends
Avoided pumping energy cost	Operational saving · canal operator	Yes
Energy sold to the market	Revenue · project vehicle	Yes
Cooling-effect PV yield uplift	Performance · project vehicle	Yes



The adaptation value is exactly where no cash flows.

Structuring logic: energy revenue carries the debt; the measured water benefit justifies the concessional or public tranche that makes the whole affordable.

Bankability by retrofit vs by design

Conventional path

Bankability-by-Design

WHEN FINANCE ENTERS

At the end, once the project is already designed

At concept stage, as a design input with gates

CO-BENEFITS

Narrative text, unscored

Quantified in tiers and scored (CCBV, PDET)

PROJECT DATA

Rebuilt for every funder

One Project Digital Model, many investment cases

STAKEHOLDERS

Consulted late, if at all

Co-created with authorities, users and investors

AFTER FINANCIAL CLOSE

A one-off report

Continuous, predictive assessment against commitments

Backup: the full BbD Framework



BbD Framework

Design principles



Economic value

Avoided losses, cost savings and risk reduction in monetisable terms · IRR, NPV, DSCR



Environmental value

Biodiversity, water, carbon, ecosystem services · EU Taxonomy & DNSH · via LRI and EQI



Social value

Public health, community resilience, equitable access, local jobs · via RWI



Governance value

Institutional trust, replicability, regulatory alignment, transparency · via CGAI

Framework pillars



Project lifecycle

- Concept → feasibility → structuring → financing → implementation → operations
- Entry and exit criteria per phase



Data model

- Project Digital Model: one data layer for multi-dimensional value
- Financials, context, theory of change, lifespan events



Operational workflow

- 4-step iterative loop
- Climate scenario testing at every step



Support toolkit

- Assessment and scoring (CCBV, PDET)
- Reporting, due diligence, structuring and monitoring tools



Unified communication

- Common terminology and disclosure formats
- ESG and EU Taxonomy language from the start



Reference & replication

- Worked examples from 4 CLIMINVEST pilots
- Replication packages for portfolio financing

Pre-screening

Co-creation

Co-benefit valuation

Business model & structuring

Investor appraisal

Continuous assessment

Lifecycle phases

3 progressive levels:

- L1 capture the idea
- L2 structured development
- L3 bankability report / investment-ready dossier

Structured engagement: financial institutions, developers, public sector, technology providers
Workshops, focus groups, co-design

CCBV in 3 tiers:

- T1 direct quantification
- T2 indices: RWI, LRI, CGAI
- T3 ESG Quality Index (EQI) → PDET scorecard

Archetypes: service fees, PPA, cost savings, carbon credits
Revenue stacking
EU Taxonomy alignment

Due diligence: technical, financial, legal, ESG
Term sheet
Credit committee

Indicator tracking
Climate scenario stress tests
Realised PDET
Taxonomy & DNSH monitoring

Iterative reassessment · preceded by readiness (Academy, Marketplace); replication feeds new projects back into the Marketplace

The BbD lifecycle at a glance



BbD Framework

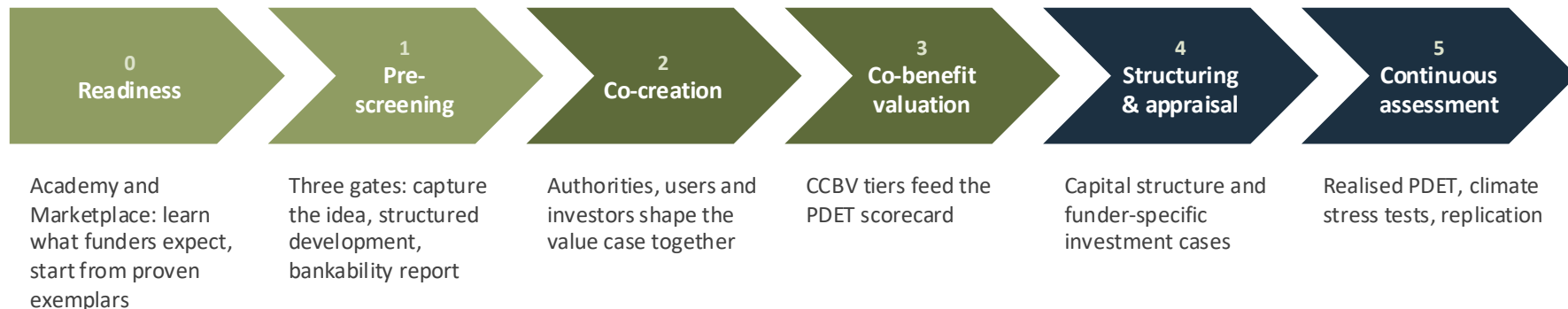
DESIGNED FOR

Economic value

Environmental value

Social value

Governance value



Iterative reassessment: each phase has entry and exit criteria, and a project steps back when the evidence changes.

Making co-benefits investable

From water saved, local jobs and resilience to indicators an investor can read, compare and price.



What do CCBV & the PDET scorecard solve ?

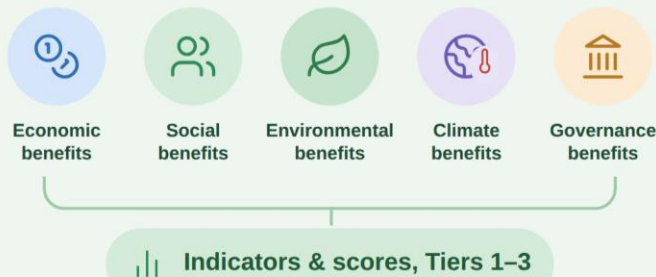
Together they close two key gaps in climate adaptation projects:

1

CCBV · CO-BENEFIT VALUATION

Benefits are hard to quantify

The CCBV method turns a project's economic, social, environmental, climate and governance benefits into **measurable indicators and scores**, in three tiers.



2

PDET · PROJECT DEVELOPMENT EVALUATION TOOL

No integrated view of the project

The PDET scorecard gives the investor and the Project Owner **a single view of the project's value**: financial performance alongside social, environmental, climate and governance benefits, weighted by the investor's priorities.


Investor /
Project Owner


One integrated, investor-weighted view



Together, CCBV and the PDET scorecard make the wider value of a climate adaptation project measurable and easy to understand for investors and Project Owners.

CCBV: three tiers, from measured to judged

CLIMINVEST Co-Benefit Valuation turns co-benefits into indicators an investor can read.

TIER 1 Direct quantification

HOW

Standard units: kWh, m³, €, tCO₂e



WHAT IT COVERS

Financial performance and core climate outputs

Energy yield · water saved · IRR

TIER 2 Proxy indices

HOW

Structured metrics, normalised to 0–100



WHAT IT COVERS

Social, resilience and governance value

RWI · LRI · CGAI

TIER 3 Expert judgement

HOW

Low / Medium / High, with mandatory evidence



WHAT IT COVERS

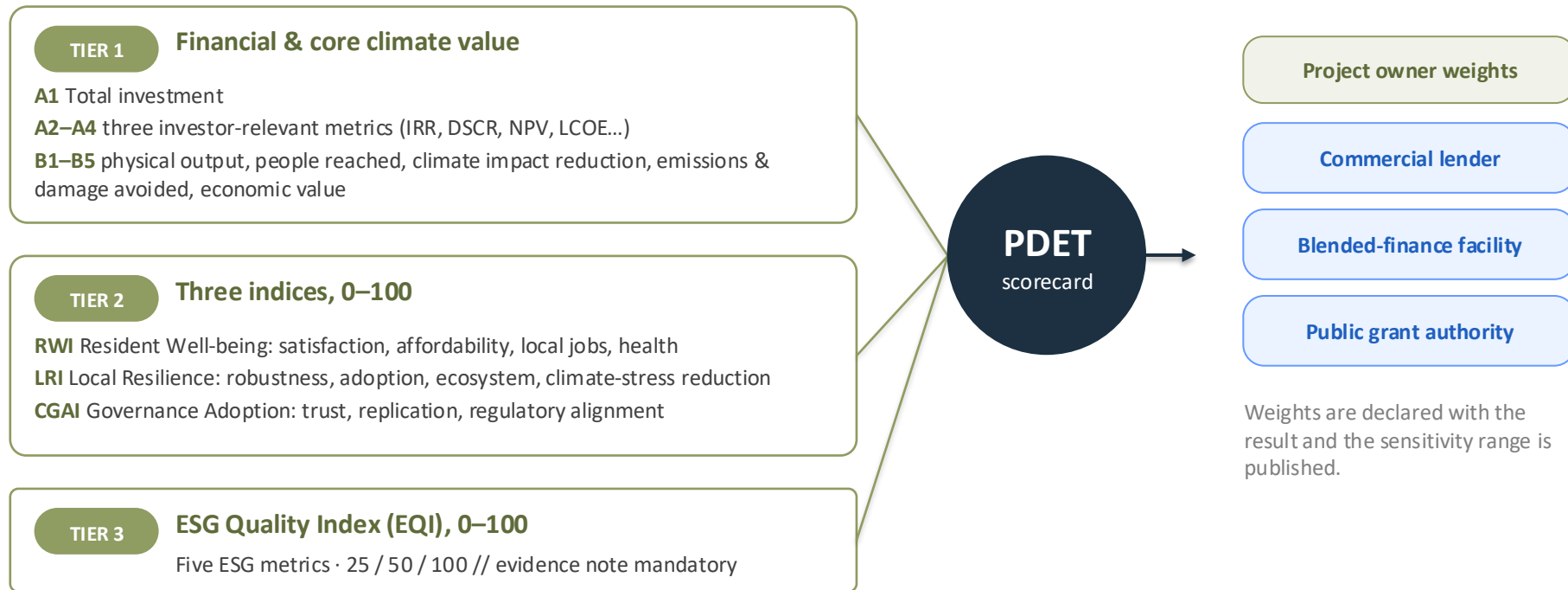
Emerging and intangible co-benefits

No evidence, no score

More precise → broader coverage. Each tier is used only where the one above it cannot measure.

The PDET scorecard: one legible profile

PDET = Project Development Evaluation Tool, the final output of the CCBV



The PDET scorecard: Project Owner example

Project Owner Composite Score – Calculation Process



The PDET scorecard: Investor example

INVESTOR-WEIGHTED COMPOSITE SCORE – CALCULATION PROCESS

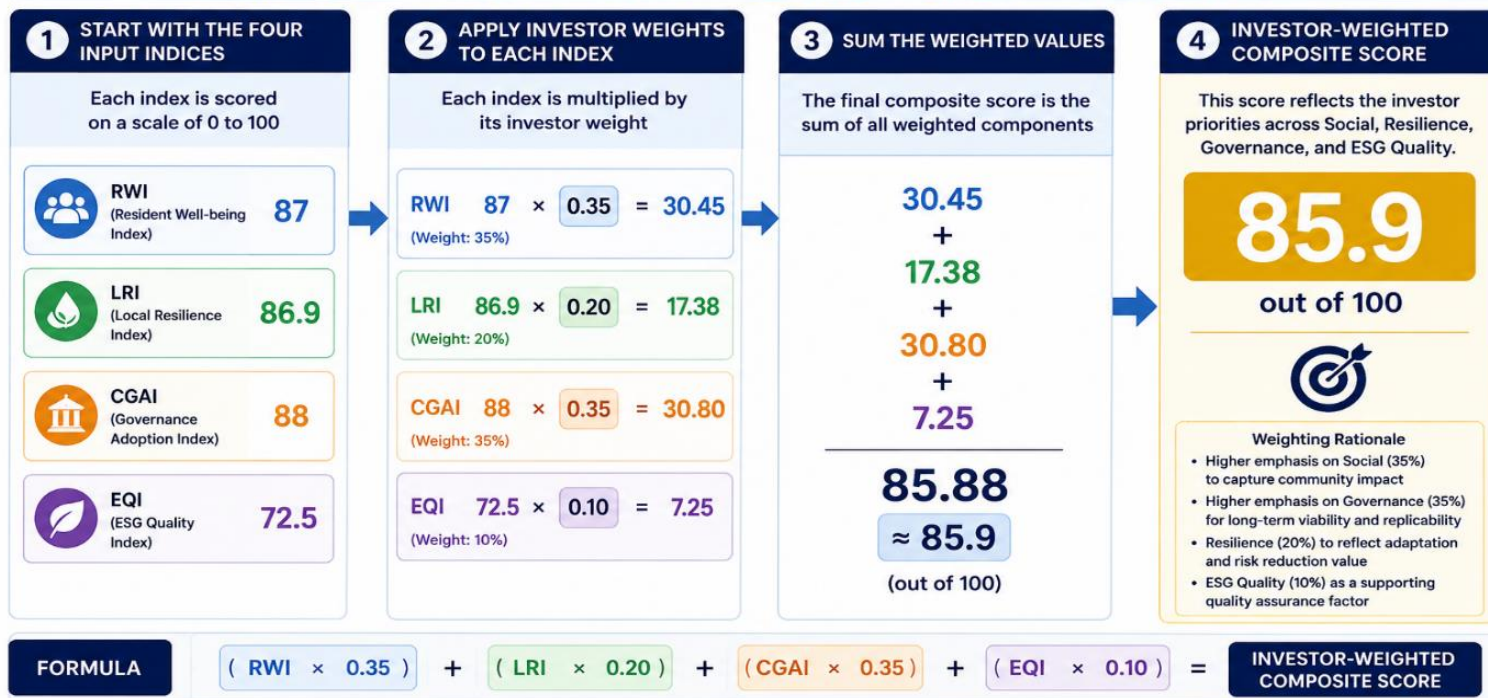
The investor composite score reflects strategic priorities using the following weights:

RWI (Social): 35%

LRI (Resilience): 20%

CGAI (Governance): 35%

EQI (ESG Quality): 10%



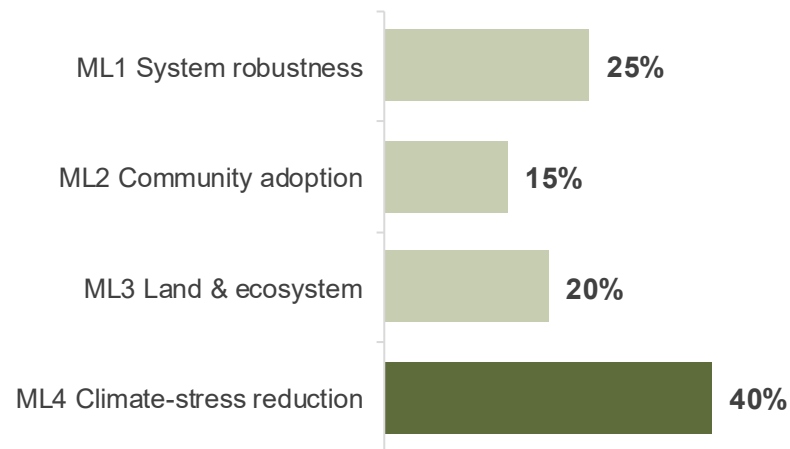
$$\text{Composite Score (Investor-Weighted)} = (RWI \times 0.35) + (LRI \times 0.20) + (CGAI \times 0.35) + (EQI \times 0.10) = 85.88 \approx 85.9$$

What makes a score **credible**

- 1 Every defined metric carries a weight
- 2 **The primary adaptation output carries the highest weight**
- 3 Weights are typology-specific and declared
- 4 Percentages are normalised against agreed ceilings
- 5 **Every score carries an evidence reference**
- 6 The composite weighting belongs to the investor

Local Resilience Index: metric weights

Water–energy typology (SOLVATE)



Weights are not fixed: each user sets them, and they are published with the score.

Committed vs Realised: the delivery ratio



Committed PDET

ex-ante

Input: design-stage commitments

Evidence: a document behind every score

Purpose: design validation, financing decision

Timing: before implementation



same metrics
same weights



Realised PDET

ex-post

Input: measured or audited outcomes

Evidence: metered data, payroll, audits

Purpose: performance verification, benchmarking

Timing: periodically, over the project's life

$$\text{Delivery ratio} = \text{Realised PDET} \div \text{Committed PDET}$$

*“A project that commits high and delivers low is a different credit proposition from one that commits modestly and delivers fully.
No single-point score tells them apart.”*

From score to finance: **three mechanisms**

A score matters only if a financing decision changes because of it. Three candidates, **tested with investors**:



MECHANISM A

Concessionality laddering

PDET bands set how much of the capital stack justifies concessional terms, pricing the value the market cannot capture.

For: **Blended-finance facilities**



MECHANISM B

KPI-linked margin ratchet

Selected metrics become KPIs in a sustainability-linked instrument; the margin moves with the Realised score.

For: **Sustainability-linked lenders**



MECHANISM C

Lower due-diligence cost

The evidenced PDET package replaces part of the lender's own ESG assessment, cutting transaction cost.

For: **Commercial lenders**

Quick poll: where would an evidenced scorecard make a real difference for you? A, B, C

slido

If a project came to you with an evidenced PDET scorecard, where would it make a real difference?

Options:

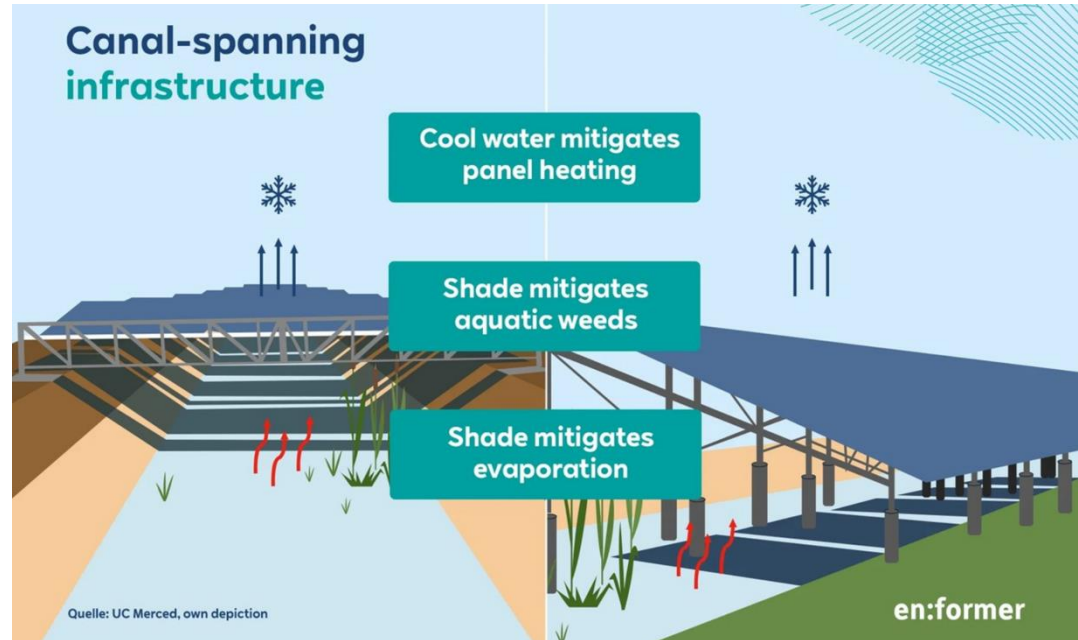
- A: Sizing the concessional money in a blended deal
- B: Linking the loan margin to delivered results
- C: Cutting due-diligence cost

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Pilot 1: SOLVATE

Solar Overlay for Water Adaptation, Transformation and Energy

-  PV panels over an open irrigation canal at Tavropos / Lake Plastira, Thessaly (GR)
-  Cuts evaporation; powers pumping and sells surplus energy
-  A water–energy model under SSP climate pathways is the assessment engine itself
-  Designed, permitted and investor-backed; awaiting grid-connection terms



Two journeys on one model: the authority explores water security; the investor compares returns under each climate scenario.

Same intervention, different bankability

PILOT**Tavropos, Thessaly (GR)****DOWNSTREAM**

agricultural irrigation users

COUNTERPARTY

no creditworthy power buyer

STRUCTURE

blended: water value justifies the concessional tranche

REPLICATOR**Narcea Canal, Asturias (ES)****DOWNSTREAM**

industrial off-taker & regional public water utility

COUNTERPARTY

creditworthy PPA counterparty

STRUCTURE

energy revenue can carry PPA-backed debt

Transfers unchanged

- ✓ Model structure
- ✓ water–energy logic
- ✓ climate-forcing method
- ✓ PDET metrics & weights
- ✓ pre-screening
- ✓ report structure

Must be localised

- Institutions
- Demand
- Regulation
- Revenue model

Bankability depends on who is downstream of the intervention, not only on the technology.

From framework to tools: where AI comes in

BBD STEP	AI CAPABILITY IN CLIMINVEST	TODAY
 Readiness	Learning Centre & Marketplace learn what funders expect; start from proven blueprints	12:25
 Pre-screening & data capture	Information extraction feasibility studies, permits and models populate the Project Digital Model	10:35
 Valuation & due diligence	Financial question answering investors query the evidence behind every score	11:25
 Investor appraisal	Bankability report generation one digital model, funder-specific investment cases	11:45
 The whole workflow	CLIMINVEST Studio (demo) the BbD phases run as one guided process	12:05

AI drafts, evidence decides: every generated figure must trace back to a scored, evidenced input.

How to get involved

The framework is being validated with pilots and investors right now. Three ways to shape it:



Lenders & investors

Set real weighting profiles, and tell us which finance mechanism would change your decision.

Investor co-creation track



Fintechs & data providers

Connect your data and tools to the Project Digital Model and the CLIMINVEST Studio.

Integration & data



Developers & public bodies

Run a project through pre-screening, from first idea to bankability report.

Pre-screening

Talk to me today, or write to mygdakos@bspoke.gr or reach us through www.climinvest-horizon.eu



Thank you

Questions and discussion



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